



# Knowledge transfer accelerated turbine blade optimization via an sample-weighted variational autoencoder

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## ABSTRACT

The aerodynamic shape optimization (ASO) of turbine blades is a typical expensive black-box problem. With the ever increasingly strict design requirements, high-fidelity simulations are demanded, and the related simulation cost increases sharply, which means fewer simulation calls are allowed within a design optimization cycle. Then, the problem as “how to achieve better solution with fewer simulations” becomes rather crucial. Inspired by the concept of transfer learning, we propose to leverage the samples of related completed tasks (also known as source samples) to address the above problem. Specifically, an sample-weighted variational autoencoder (VAE) is proposed, which includes the source samples as part of the training dataset to boost the proportion of high-scoring solution candidates in the learnt design space of target problem, and thus increasing the probability of attaining substantially better solutions. The including of source samples in VAE training also helps to increase the accuracy of reparameterizing the source samples in target design space. Furthermore, a multi-fidelity surrogate is used to guide the optimization search, which extracts information from the source samples to help the algorithm to more quickly arrive at the small neighborhood of the true optimal of the target problem. With the above, a knowledge transfer accelerated aerodynamic shape optimization procedure is proposed for the turbine blade design, labeled as KT-ASO. Through test on a highly-loaded low-pressure turbine blade and compared against the ASO procedures without knowledge transfer, the proposed KT-ASO is shown to reduce at least 50% function calls when achieving similar optimal solutions. In the meantime, under the same sample budget, our proposed KT-ASO is shown to attain much better solutions than the compared ASO procedures.

## 1. Introduction

The aerodynamic shape optimization (ASO) is an everlasting hotspot for its pivotal role in the design process of turbomachinery cascades [1–4], etc. Generally, the ASO is carried out as follows. First, a parameterized design space  $\mathcal{Z}$  is built by using the techniques such as Free-Form Deformation (FFD) [5,6] or generative model based methods [7–9]. Then, an optimization is carried out over the parameterized design space  $\mathcal{Z}$ . Since the performance evaluation by CFD can be quite expensive, surrogate-based optimization (SBO) algorithms [10,11] are frequently used for the ASO process, which build approximations to replace part of the CFD simulations in the optimization searching process. Nonetheless, the number of CFD evaluations required for SBO algorithms can increase exponentially with the increase of design space dimensionality [12]. More importantly, in order to meet the increasingly strict design requirements, more advanced CFD techniques with higher-

fidelity simulations have to be used, but the related computational cost of single run of CFD evaluation increases dramatically [13–15]. It means, the number of CFD evaluations that is allowed within a design cycle can be reduced sharply for more and more elaborated turbine design. In such situations, the issue as “how to achieve even better solutions without increasing or even reducing the number of CFD evaluations” becomes rather crucial.

To resolve the above issue, the dimensionality reduction (DR) techniques [16,17] are frequently studied, which seek to accelerate the optimization progress by searching only those dimensions that can influence the aerodynamic performance significantly. For instance, Chen et al. [18] proposed Bézier-GAN which selects three latent codes out of the 13 input parameters to capture the major shape variability of the airfoil design space. And consequently, the number of CFD evaluations needed for the related ASO process with the three selected latent variables can be reduced by over an order of magnitude, when com-

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## Nomenclature

|               |                                                                |                        |                                                           |
|---------------|----------------------------------------------------------------|------------------------|-----------------------------------------------------------|
| $C_{pt}$      | total pressure loss coefficient                                | $\mathcal{Z}$          | parameterized aerodynamic design space                    |
| $C_{ax}$      | axial chord length                                             | $\rho$                 | correlation coefficient in multi-fidelity surrogate model |
| $L$           | loss function of VAE model                                     | $\lambda_1, \lambda_2$ | weight coefficients of VAE loss function                  |
| $m$           | mass rate                                                      | <i>Subscripts</i>      |                                                           |
| $t$           | blade pitch                                                    | S                      | source task                                               |
| $\mathbf{x}$  | an aerodynamic shape sample represented by scatter coordinates | T                      | target task                                               |
| $\mathcal{X}$ | scatter coordinates based design space                         | ref                    | reference design                                          |
| $\mathbf{z}$  | an sample in the parameterized design space                    |                        |                                                           |

paring to the conventional ASO procedure. However, in their further study, they also showed that while dramatically reducing the number of CFD evaluations, the final optimal solutions that are obtained with only three selected latent codes are worse than those obtained with the full input parameters of Bézier-GAN. The reason behind is explained as follows. That is, while the selected three latent codes can capture the major shape variability of the airfoil design space, the remaining 10 input parameters of Bézier-GAN can help to generate airfoils with even better performance. Such drawback is shared with most DR-based optimization procedures [19,20]. Different from DR methods, Yonekura et al. [21] bring in the concept of reinforcement learning (RL) to accelerate the aerodynamic design process of turbine airfoils and thus addressing the above issue. According to their observations, while the computational time for training the RL model is long, the trained RL model can be reused to solve different ASO problems, and the computational time to solve the ASO problem by the trained RL model is very short. Similar observation is also seen in the work of Li et al. [22]. In other words, in the long run, the RL method can be a promising approach to resolve the aforementioned issue.

In addition to the DR methods and RL approach, the concept of transfer learning (TL) [23] is proposed in recent years, which may provide a timely solution to obtain better optimized results without increasing or even reducing the number of CFD evaluations. Specifically, inspired by the fact that the real-world problems are seldom in isolation and we humans can often make use of the past experiences when solving a new task, the basic idea of TL is to quickly arrive at the small neighborhood of the true optimal of the target problem, by taking full advantage of the similarities between the target and the related completed tasks [24–28]. The TL has also drawn wide attention for solving expensive optimization problems, known as transfer optimization (TO) [29], and a series of success stories has been reported in this relatively young field [30–34].

When applying TO to accelerate the turbine design, the following two problems need to be considered. First, how to build a shared design space between the target and the related completed task, or how to accurately reparameterize the samples of the completed tasks (also known as source samples) into the target design space. Upon that, the second problem is how to efficiently extract useful information from the source samples to accelerate the target design process. Thus far, the previous studies mainly focus on the second problem, i.e., by assuming that the target and the related completed task have already been put into a shared design space, they focus on developing knowledge transfer strategies that can effectively leverage source samples to accelerate the convergence of an optimization algorithm for the target problem. Differently, the first problem is also addressed in this paper, and accordingly a multi-fidelity surrogate (MFS) based TO algorithm is proposed to address the second problem.

Specifically, to make TO applicable for turbine design, the variational autoencoder (VAE) [35] is used to build the shared design space, which can accurately reparameterize the source samples into the target design space. In particular, different from the previous VAE-based aerodynamic design studies [36], much attention is paid to the training

scheme of VAE. More specifically, an sample-weighted training strategy is proposed, which makes use of the source samples to train the VAE-based parameterization space. As can be seen in the testing analysis in the following sections, such training strategy helps to improve the reparameterization accuracy of source samples in the target design space. More importantly, the proposed training strategy also helps to boost the proportion of high-scoring solution candidates in target design space [37], and thus making it easier to attain optimal solutions in the target task. After that, an MFS-based TO algorithm is proposed by treating the source samples as low-fidelity (LF) samples while the target samples as high-fidelity (HF) samples. The related knowledge transfer accelerated framework for the aerodynamic shape design of turbine blades is labeled as KT-ASO, which is demonstrated through shape optimization of an ultra-highly loaded turbine blade.

The remainder of this paper is organized as follows: the basics of standard ASO procedure, generative model based shape optimization, transfer optimization and Bayesian optimization are briefly introduced in Section 2. Then, the proposed sample-weighted VAE for blade parameterization is illustrated in Section 3. After that, the knowledge transfer accelerated blade optimization framework, namely KT-ASO, is presented. Next, to verify the effectiveness of proposed KT-ASO, an aerodynamic shape optimization of highly-loaded turbine blade is carried out in Section 5. And finally, some conclusions are drawn in Section 6.

## 2. Related works

### 2.1. Standard ASO procedure and geometric parameterization method

The general procedure of standard ASO is as follows (see also Fig. 1). First, a reference design  $\mathbf{x}_{ref}$  that is represented by hundreds of scatter coordinate points is selected. Second, the techniques such as geometric parameterization method, FFD and etc., are used to fit the blade profile of the reference design. And accordingly, several or dozens of design variables are selected to build the parameterized design space  $\mathcal{Z}$ , of which blade profiles are adjusted around  $\mathbf{x}_{ref}$ . Then, the optimization is carried out over  $\mathcal{Z}$ , with the following equation:

$$\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) = \min_{\mathbf{z} \in \mathcal{Z}} f(P_\theta(\mathbf{z})) = \min_{\mathbf{z} \in \mathcal{Z}} y(\mathbf{z}) \quad (1)$$

where,  $\mathbf{x}$  is the blade profile that is represented by scatter coordinates, and  $\mathcal{X}$  is the related design space.  $P_\theta(\cdot)$  is the parameterization model built by Pritchard's geometric parameterization method and etc.,  $\mathbf{z}$  is the design variable vector of the related parameterization method.

More specifically, the geometric parameterization method is presented in Fig. 2. It consists of 11 geometric parameters, such as the inlet geometric angle  $\beta_1$ , upper wedge angle  $\epsilon_{up}$ , outlet airflow angle  $\beta_2$ , outlet deflection angle  $\delta_{out}$ , angle of circle center line  $\gamma$ , axial chord length  $C_{ax}$ , etc. In the meantime,  $o$  measure the throat size, and  $t$  is blade pitch. More details of the geometric method can be seen in [38].

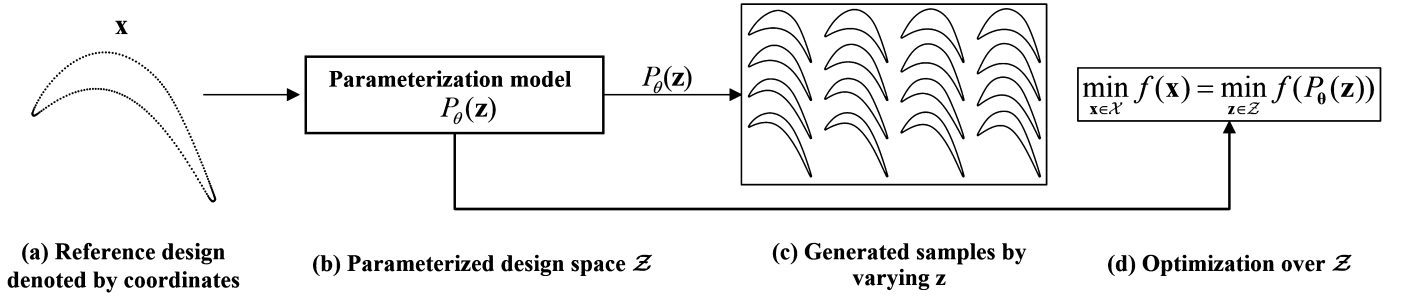


Fig. 1. Standard procedure of conventional aerodynamic shape optimization.

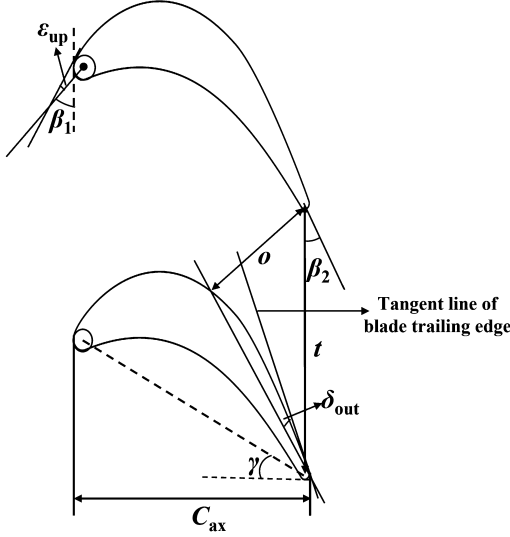


Fig. 2. Geometric parameterization method of turbine blade profile.

## 2.2. Generative model based shape optimization

In addition to the standard ASO procedures that are shown in Fig. 1, the generative model based shape optimization (GMO) method is proposed, which has drawn wide attentions in recent years. Specifically, unlike the conventional parameterization methods that build the design space based on a single reference design, the deep learning models are used in GMO to learn the latent design features of the blade profiles, which have been shown to form the design space with fewer variables but have much better shape variability. Moreover, the ratio of abnormal samples in the learnt design space of GMO is also significantly reduced [7–9].

The general process of GMO is illustrated in Fig. 3. First, a dataset  $X$  that contains a large number of aerodynamic shape samples is collected. Usually, each sample in  $X$  is represented by hundreds of scatter coordinates in a uniformed format, denoted by  $\mathbf{x}$ , and the related design space is labeled as  $\mathcal{X}$ . Second, a generative model such as the generative adversarial net (GAN), variational autoencoder (VAE), etc. is used, which is trained to encode the samples of  $\mathcal{X}$  into a latent design space  $\mathcal{Z}$  that is represented by dozens of variables. Then, the aerodynamic optimization is carried out over the latent design space  $\mathcal{Z}$ , with the following equation:

$$\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) = \min_{\mathbf{z} \in \mathcal{Z}} f(G_\theta(\mathbf{z})) = \min_{\mathbf{z} \in \mathcal{Z}} y(\mathbf{z}) \quad (2)$$

where,  $G_\theta$  is the trained generative model, and  $\mathbf{z}$  is a sample vector in the latent space  $\mathcal{Z}$ .

While a series of success stories have been reported, Tripp et al. [37] pointed out the potential “failure mode” of GMO. That is, a large portion of training samples of  $G_\theta$  are the low-scoring points with their objective function values far from optimal, as they are generated ran-

domly without objective function evaluations. Then, most regions of the learnt latent space of  $G_\theta$  are low-scoring points but leave insufficient space for high-scoring points that lie outside the training set. As a consequence, even a perfect optimization algorithm with unlimited function calls might be unable to find a solution that is substantially better than the best point in the original dataset, simply because such a point may not exist in the feasible region. However, the strategies on how to improve the sample distributions of the training set of  $G_\theta$  are rarely discussed in the previous studies, which is addressed from the view of transfer optimization in this paper.

## 2.3. Basics of transfer optimization

The mathematical expression of transfer optimization is formulated as follows:

$$\begin{aligned} \min \{T | S_1, \dots, S_k, K(i)\} \\ T = \min_{\mathbf{z} \in \mathcal{Z}} y(\mathbf{z}) \\ S_k = \min_{\mathbf{z}_k \in \mathcal{Z}_k^{(S)}} y(\mathbf{z}_k) \end{aligned} \quad (3)$$

where,  $T$  denotes the target optimization problem,  $S_1, \dots, S_k$  are the solutions of related completed tasks,  $K(i)$  is the knowledge transfer strategy that can effectively gain information from  $S_1, \dots, S_k$  at  $i^{th}$  optimization cycle, and  $\mathcal{Z}$  and  $\mathcal{Z}_k^{(S)}$  are the parameterized design space of the target and completed tasks, respectively.

As shown in Fig. 4, the similarity between tasks means that the optimal solutions of related completed tasks and the target problem are probably located close to each other. Then, the basic idea of TO is to extract information from  $S_1, \dots, S_k$  to more quickly arrive at the vicinity of optimal solution  $\mathbf{z}^*$  of  $T$ . To achieve this goal, the previous studies focus on building  $K(i)$  in an optimization algorithm, by presuming that the source samples of related completed tasks have already been put into the design space of target problem.

Differently, to make TO more applicable for the shape optimization of turbine blades, much attention is paid to the reparameterization of  $S_1, \dots, S_k$  into  $\mathcal{Z}$  by using VAE in this paper. Moreover, to address the “failure mode” of GMO that is discussed in Section 2.1, a sample-weighted training scheme is proposed, which makes use of  $S_1, \dots, S_k$  to improve the sample distributions of the training set for VAE. Consequently, the proportion of high-scoring points in learnt latent space  $\mathcal{Z}$  can be greatly increased, which helps to more easily attain substantially better solutions in the target design space. From this view, the proposed training scheme of VAE is a novel form of  $K(i)$  to facilitate the optimization of target problem.

## 2.4. Bayesian optimization

Bayesian optimization (BO) [39,40] is well-known to be sample-efficient, which is most frequently used for solving expensive black-box problems. The general procedure of BO is as follows. First, a surrogate of the objective function is built by using Gaussian process (GP). Second, guided by GP surrogate, an acquisition function is used to select

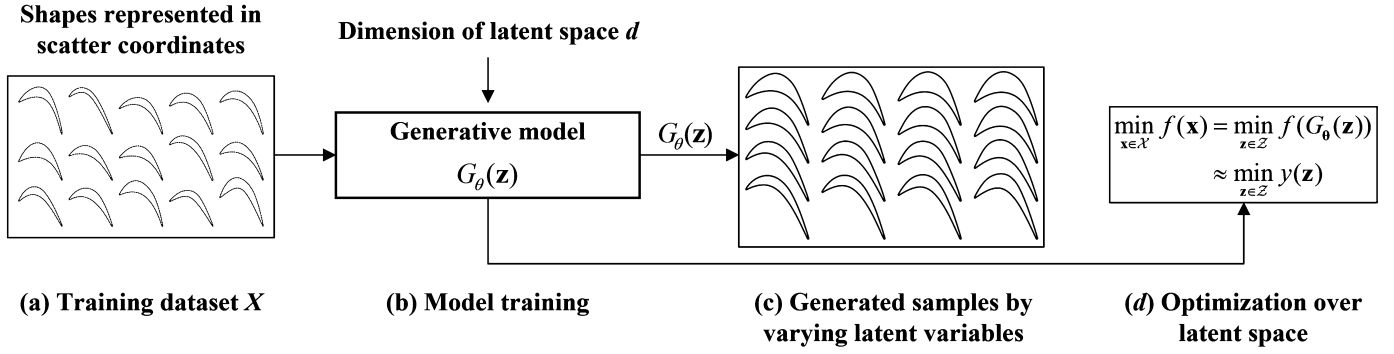


Fig. 3. Schematic of generative model based aerodynamic shape optimization.

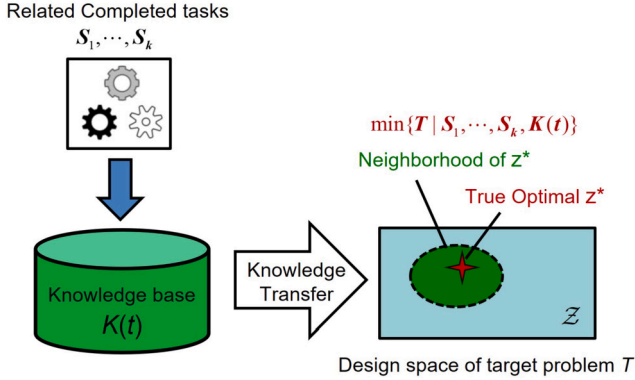


Fig. 4. Scheme of transfer optimization.

the next promising sample to query. Third, the new queried sample is added to the training set to update the GP surrogate. The algorithm repeats the above process until the termination condition is met.

As a key component of BO, GP formulates the function prediction  $Y(\mathbf{x})$  as a Gaussian random variable with the following equation:

$$Y(\mathbf{x}) \sim \text{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \quad (4)$$

where  $m(\mathbf{x})$  denotes the mean function, and  $k(\mathbf{x}, \mathbf{x}')$  is the covariance function between samples  $\mathbf{x}$  and  $\mathbf{x}'$ . Usually, we take  $m(\mathbf{x}) \equiv 0$  and use the popular squared exponential (SE) covariance function to describe the covariance correlations. Accordingly, the function prediction and related mean squared error of the prediction are formulated as follows:

$$\hat{y}(\mathbf{x}) = \mathbf{k}(\mathbf{x}, X)(K + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} \quad (5)$$

$$\hat{\sigma}^2(\hat{y}(\mathbf{x})) = \mathbf{k}(\mathbf{x}, \mathbf{x}) - \mathbf{k}(\mathbf{x}, X)(K + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}(X, \mathbf{x}) + \sigma_n^2$$

where,  $\sigma_n^2$  is the Gaussian noise,  $K$  is the covariance matrix of the input variables, and  $\mathbf{k}$  is the cross-correlation vector between  $\mathbf{x}$  and  $X$ .

As another key component of BO, the acquisition function helps the algorithm to select the next promising sample to query. The function of expected improvement (EI) [41] is the most frequently used acquisition function. Assuming that the function prediction  $Y(\mathbf{x})$  comes from a GP as  $Y(\mathbf{x}) \sim N(\hat{y}(\mathbf{x}), \sigma^2(\mathbf{x}))$ , the basic idea of EI is to measure how much objective improvement can be attained with respect to the current best solution  $y_{\min}^{(t)}$ , with the following equation:

$$EI(\mathbf{x}) = (y_{\min}^{(t)} - \hat{y}(\mathbf{x}))\Phi(u) + \sigma(\mathbf{x})\phi(u) \quad (6)$$

$$u = (y_{\min}^{(t)} - \hat{y}(\mathbf{x})) / \sigma(\mathbf{x})$$

where,  $\Phi(\cdot)$  and  $\phi(\cdot)$  denote the standard normal distribution function and density function, respectively. In each optimization cycle, the next sample to query is selected by maximizing the EI function as follows:

$$\mathbf{x}^* = \arg \max_{\mathbf{x}} EI(\mathbf{x}) \quad (7)$$

### 3. Sample-weighted VAE for blade parameterization by including source samples

In this section, the basics of VAE for blade parameterization is first introduced. Then, our proposed sample-weighted VAE is discussed in detail.

#### 3.1. Basics of VAE for blade parameterization

The basic architecture of VAE for blade parameterization is shown in Fig. 5, which consists of two parts, i.e. encoder and decoder. The function of decoder is to transform a sample  $\mathbf{x}$  from the scatter coordinates based design space  $\mathcal{X}$  to the much lower-dimensional latent space  $\mathcal{Z}$ , i.e., building the mapping as  $\mathbf{x} \rightarrow \mathbf{z}$ . Inversely, the decoder builds the mapping as  $\mathbf{z} \rightarrow \mathbf{x}$ . The basic training process of VAE for blade shape parameterization is as follows. First, a set of turbine blade samples  $X$  that are represented by scatter coordinates in a uniformed format is collected, by using the sampling techniques such as Latin hypercube sampling, etc. Second, determine the dimension of latent space of VAE, and design the neuron network architecture of the encoder and decoder. Third, train the VAE model with  $X$ . Specifically, in each epoch of the training process, the sample  $\mathbf{x}$  is first fed into the encoder to obtain the related latent representation  $\mathbf{z}$ . Then, a reconstructed  $\hat{\mathbf{x}}$  is obtained by feeding  $\mathbf{z}$  to the decoder. After that, the weights of neurons of encoder and decoder are adjusted accordingly by calculating the difference between  $\mathbf{x}$  and  $\hat{\mathbf{x}}$ . The above process is repeated until the termination condition is met.

Additionally, the training loss of VAE model for blade shape parameterization is shown as follows:

$$L(\mathbf{x}) = KL(\mathbf{x}) + \lambda_1 MSE(\mathbf{x}) + \lambda_2 L_P(\mathbf{x}) \quad (8)$$

where,  $KL$  is short for the Kullback-Leibler divergence [42], which measures the sample variability of the design space.  $MSE$  is short for the mean squared error, which measures the similarity between the training sample  $\mathbf{x}$  and reconstructed sample  $\hat{\mathbf{x}}$ .  $L_P$  is the regularization term that considers the specific geometric constraints of turbine blade, which will be discussed in detail in Section 5 when discussing the engineering case.  $\lambda_1$  and  $\lambda_2$  are the coefficients that are used to adjust the weights of different losses in the training process.

After training, various blade profiles can be generated by feeding the related latent representation  $\mathbf{z}$  to the VAE decoder, and aerodynamic optimization of turbine blades can be carried out over the latent space  $\mathcal{Z}$ . In the meantime, by first representing the blade profile in a uniformed format of scatter points as  $\mathbf{x}$  and then feeding it to the VAE encoder, the samples of related completed tasks can be reparameterized into the latent space to enable transfer optimization of the target turbine design.



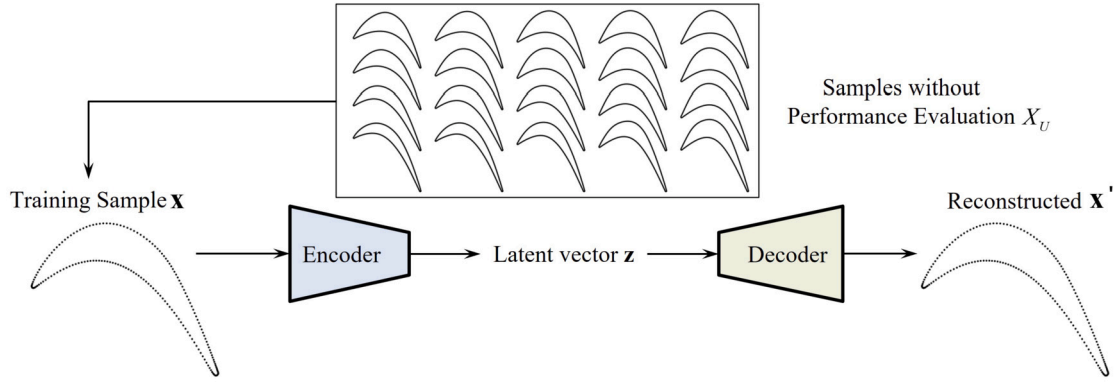


Fig. 5. Scheme of shape parameterization with basic VAE.

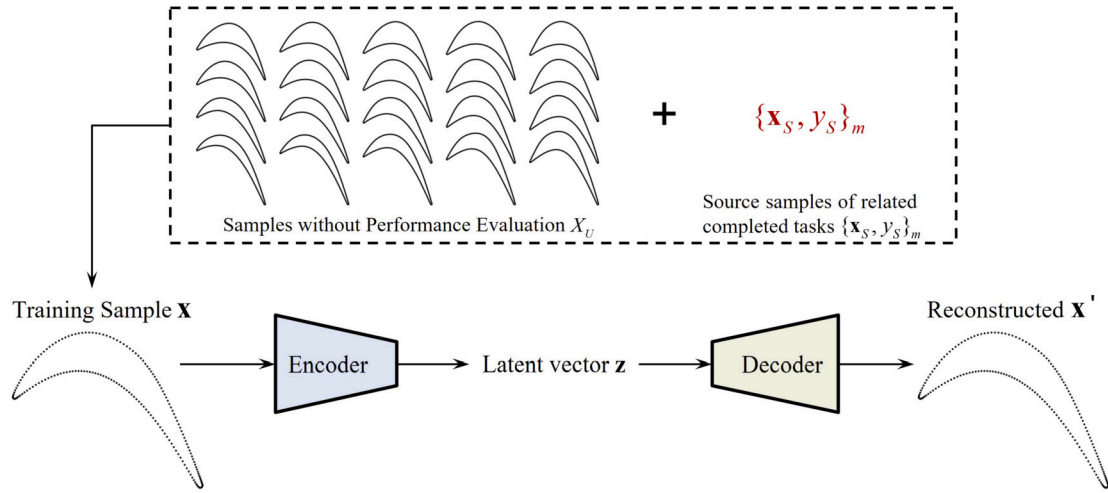


Fig. 6. Scheme of shape parameterization with sample-weighted VAE.

### 3.2. Sample-weighted training scheme

As pointed out by Tripp et al. [37], the sample distributions of the learnt design space of VAE are roughly proportional to its frequency in the training dataset. Then, if the optimized solutions of the related completed tasks (also known as source samples) lie outside the training dataset, the reparameterization of these samples to the latent space of target problem can be in poor accuracy, which may lead to the “negative transfer” in the following transfer optimization process, i.e., the use of source samples may delay rather than facilitating the target optimization progress. Hence, to address the potential “negative transfer” issue, we propose to include the source samples as part of the training dataset, as shown in Fig. 6. Additionally and more importantly, since the optimal solutions of the target and the related completed tasks are similar to each other, the including of the source samples as part of the training dataset can also help to increase the proportion of high-scoring samples in the learnt design space of VAE, which then can make it easier to attain substantially better solutions of the target problem.

Also note that thousands of training samples are often required for the effective training of VAE-based deep learning model, so except the dozens of source samples from the related completed tasks, the majority of the samples in the training dataset are the points which are randomly generated without function evaluations. In other words, the proportion of the source samples and the remaining unevaluated training points is imbalanced. Then, if we directly merge them together as a whole for the VAE training, the distinct features of the source samples can be ignored.

In view of the above imbalanced proportion issue, the sample-weighted training strategy is proposed. More specifically, in each epoch of the training process, we propose to fix the number of source samples and the unevaluated samples in a training batch, by using the following training loss:

$$L_w = \sum_{i=1}^{n^{(S)}} L(\mathbf{x}_i^{(S)}) + \sum_{j=1}^{n^{(T)}} L(\mathbf{x}_j^{(T)}) \quad (9)$$

where, the formulation of  $L(\cdot)$  is exactly the same as the definition in Eq. (8), and  $\mathbf{x}_i^{(S)}$  and  $\mathbf{x}_j^{(T)}$  are the samples that are drawn from the source sample set and the group of unevaluated samples, respectively, and the ratio as  $n^{(S)}/n^{(T)}$  in a training batch is fixed in each epoch. With the above, the training procedure of proposed sample-weighted training can be summarized as follows:

- (1) Generate unevaluated blade profiles by using the Latin hypercube sampling techniques, etc., which are represented in a uniformed format of scatter coordinates and labeled as  $X_U$ .
- (2) Gather the source samples of completed tasks, which are represented in the same format as  $X_U$  and denoted by  $X_S$ . Accordingly, the whole training dataset  $X$  is made of two parts, i.e.,  $X = [X_U, X_S]$ .
- (3) Determine the neuron network architecture of VAE including the dimension of the latent space, etc.
- (4) Train the VAE model with Eq. (9). Specifically, in each epoch of the training process, randomly drawn  $n^{(S)}$  and  $n^{(T)}$  samples from  $X_S$  and  $X_U$ , respectively, to build the training batch.

More sampling details of  $X_U$  and the effectiveness of proposed sample-weighted training scheme of VAE will be discussed in Section 5 for a turbine blade design.

#### 4. Knowledge transfer accelerated blade optimization framework

After building the design space and encoding the source samples into the learnt design space of the target problem, the multi-fidelity surrogate (MFS) is used to gain knowledge from the source samples and thus facilitating the optimization search of the target problem. And accordingly, a knowledge transfer accelerated aerodynamic shape optimization framework is presented, labeled as KT-ASO. The technical details of MFS and KT-ASO are discussed in the following paragraphs.

##### 4.1. Multi-fidelity surrogate for transfer optimization

Note that the approximation accuracy of the surrogate can greatly influence the progress of the surrogate-based optimization. However, due to the huge cost of CFD simulations, the number of target training samples particularly at the early stage of the optimization process can be too limited to build an accurate surrogate. Therefore, the source samples are used to augment the small number of target training samples to build a more accurate surrogate. More specifically, considering the fact that the objective function trend of target problem and related completed tasks are similar but cannot be the same exactly, the source samples from related completed tasks are taken as low-fidelity samples, while the evaluated samples of target optimization problem are treated as high-fidelity samples. Then, the correlations between the samples of the completed tasks and the target problem and between an observed sample  $\mathbf{z}$  and the training samples as well can be modeled by the following multi-fidelity Gaussian kernel functions:

$$K_M(Z^*) = \begin{bmatrix} K_T & \rho K_{ST} \\ \rho K_{TS} & K_S \end{bmatrix} + D, \quad k_M(\mathbf{z}, Z^*) = [k(\mathbf{z}, Z_T), \rho k(\mathbf{z}, Z_S)] \quad (10)$$

where,  $Z_T$  and  $Z_S$  are the high- and low-fidelity samples that come from the target optimization problem and related completed task, respectively, and  $Z^* = [Z_T, Z_S]$ , and  $\rho$  is the correlation coefficient that models the similarity between  $Z_T$  and  $Z_S$ . Additionally,  $D$  is a  $2 \times 2$  diagonal matrix with diagonal elements  $\{\sigma_{n,j}^2\}_{1 \leq j \leq 2}$ , and  $\sigma_{n,1}^2$  and  $\sigma_{n,2}^2$  are the noise term associated with the target and source samples, respectively. Upon that, at an unknown site  $\mathbf{z}$ , the objective function prediction  $y_M(\mathbf{z})$  and the related prediction uncertainty  $\sigma_M^2(\mathbf{z})$  can be expressed as follows:

$$\hat{y}_M(\mathbf{z}) = k_M(\mathbf{z}, Z^*) K_M^{-1} [Y_T, Y_S]^T, \quad \sigma_M^2(\mathbf{z}) = k_M(\mathbf{z}) - k_M(\mathbf{z}, Z^*) K_M^{-1} k_M(Z^*, \mathbf{z}) + \sigma_{n,1}^2 \quad (11)$$

This way, the source samples can be combined with the small number of target training samples to build an MFS of the target problem, which can have much better approximation accuracy than the surrogate that is built with target samples alone.

More intuitively, the advantage of employing samples from related tasks as low-fidelity sample to establish MFS approximation can be illustrated as Fig. 7. When only using the target samples to build the single-fidelity surrogate, obvious gap is seen when comparing it to the true function curve. Differently, when combining the target samples with transfer source samples to build a multi-fidelity surrogate, the approximation curve can perfectly fit the true function. Then, as the surrogate approximation accuracy can greatly influence the convergence rate of the surrogate-based optimization algorithms, the much better approximation accuracy of MFS can help the algorithm to arrive at the optimal solution much more quickly.

Upon building the MFS approximation, the EI illustrated in Section 2.4 is used to guide the optimization search of the target problem, which is formulated as flows:

$$EI_M(\mathbf{z}) = (y_{\min}^{(i)} - \hat{y}_M(\mathbf{z})) \Phi(u_M) + \sigma_M(\mathbf{z}) \phi(u_M), \quad u_M = (y_{\min}^{(i)} - \hat{y}_M(\mathbf{z})) / \sigma_M(\mathbf{z}) \quad (12)$$

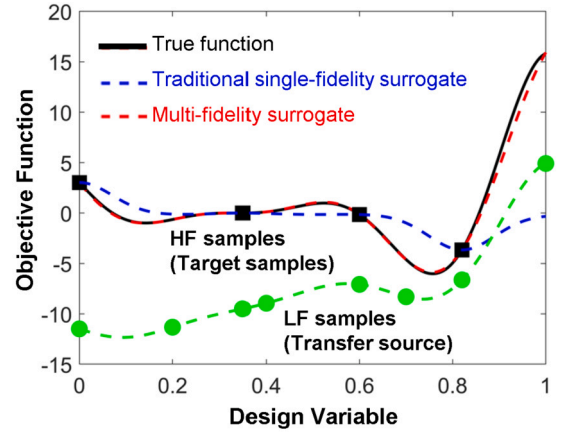


Fig. 7. Scheme of MFS approximation using transfer source samples as low-fidelity samples. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

where, the subscript  $M$  denotes the multi-fidelity surrogate, and the meaning of the other symbols can refer to the definitions in Eq. (6).

To validate the effectiveness of the proposed MFS algorithm, numerical experiment is conducted with the classical Ackley and Rosenbrock functions [43], as shown in Table 1. In order to simulate the scenarios of transfer optimization in real-world problems, the standard Bayesian optimization on source tasks is performed in advance, and the top-ranked samples are selected as the transfer source. For performance comparison, the target task is also optimized using standard single-fidelity Bayesian optimization algorithm, denoted as BO; and the proposed MFS-based optimization procedure that will be used in KT-ASO is denoted by MFO. For fair comparison, the number of initial samples is set to be 6 times of the design space dimensionality for each optimization, so as the number of transfer source samples. To reduce the effect of randomness, the performance of each algorithm is determined by 10 individual BO- and MFO-based optimizations, which have different initial target samples and transfer source samples. Fig. 8 shows the mean convergence curve and distribution of final solutions of MFO and BO over 10 runs. Obviously, the MFO is shown to have better convergence rate, and the final solutions attained by MFO also look much better than that of BO. Thereby, the effectiveness of the MFO that is used in KT-ASO to carry out optimization is well demonstrated.

##### 4.2. Framework of KT-ASO

By combining the sample-weighted VAE with multi-fidelity surrogate, the flowchart of proposed KT-ASO is shown in Fig. 9, and the detailed procedure of KT-ASO can be summarized as follows:

(1) By combining the blade profiles of source samples (i.e.,  $\{\mathbf{x}_S, y_S\}_m$ ) and the unevaluated blade profiles of the target problem (i.e.,  $\mathbf{x}_T$ ), a VAE model is trained using the sample-weighted training strategy shown in Section 3.2.

(2) By using Latin hypercube sampling (LHS), a set of target training samples are generated with the VAE decoder, which are evaluated by CFD simulations. The target samples are labeled as  $\{\mathbf{z}_T, y_T\}_t^i$ , where  $t$  denotes the  $t^{\text{th}}$  iteration of the optimization process, and  $t = 0$  at the initial stage.

(3) Using the VAE encoder, the sources  $\{\mathbf{x}_S, y_S\}_m$  are reparameterized into the latent space of the target problem, labeled as  $\{\mathbf{z}_S, y_S\}_m$ .

(4) By combining  $\{\mathbf{z}_T, y_T\}_t^i$  with  $\{\mathbf{z}_S, y_S\}_m$ , an MFGP surrogate is built (see Section 4.1).

(5) The EI acquisition function is used to select the most promising sample candidate to query, and the new sample is added in the sample set of  $\{\mathbf{z}_T, y_T\}_t^i$ .

(6) Repeat the steps of (4) and (5) until the termination condition is met.

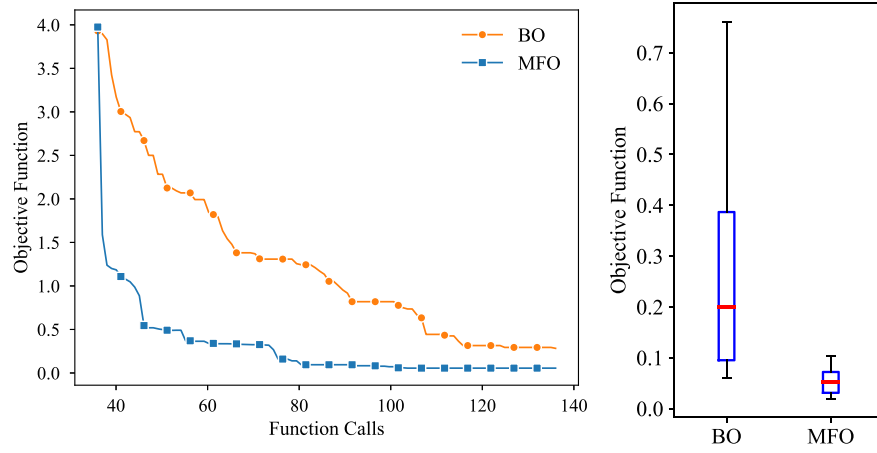


Fig. 8. The mean convergence curve and the boxplot of final solutions over 10 runs of the numerical experiment for MFS validation.

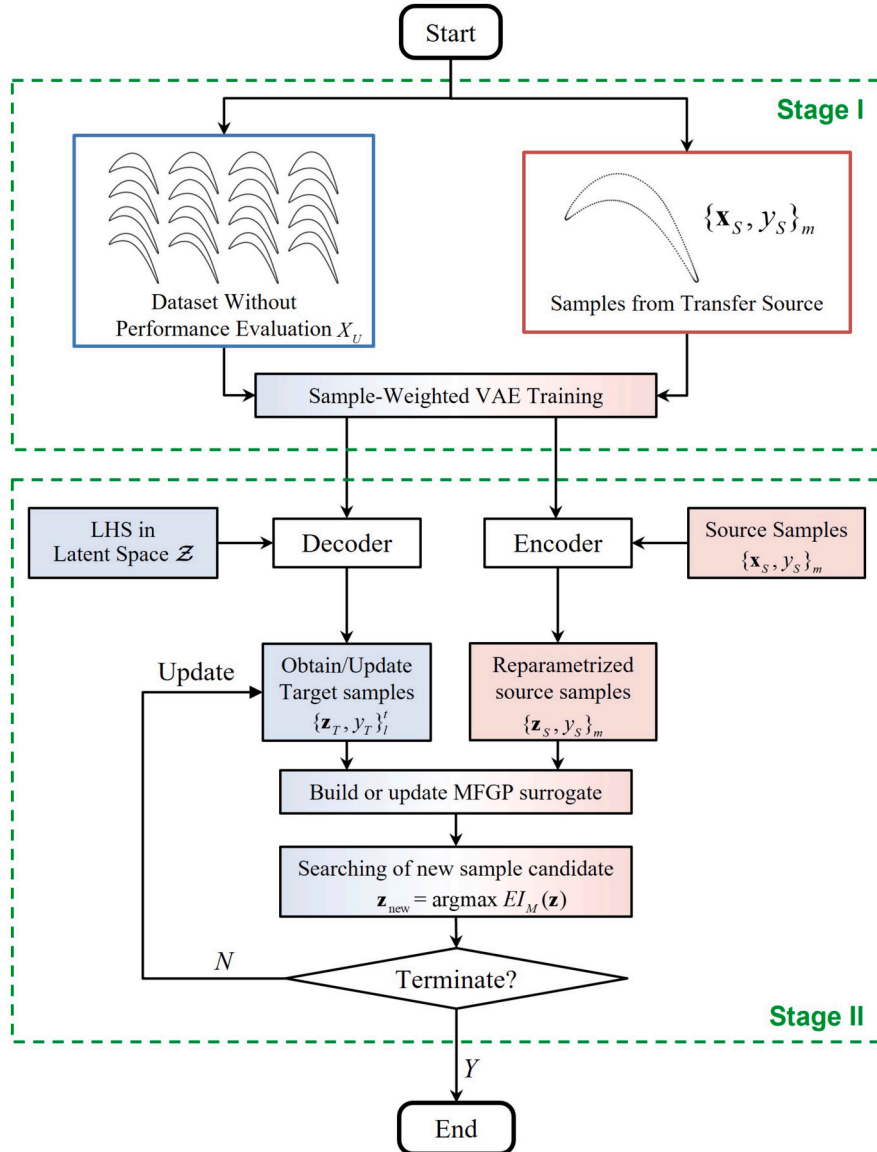


Fig. 9. Flowchart of the proposed KT-ASO procedure.

**Table 1**

Transfer source tasks and target tasks of numerical experiments for MFS validation.

|              |                                                                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source Task  | $0.9 \times \text{Ackley6d}(\mathbf{x}) + 0.1 \times \text{Rosenbrock6d}(\mathbf{x}), x_i \in [-2, 2]$                                                                  |
| Target Task  | $\text{Ackley6d}(\mathbf{x}), x_i \in [-2, 2]$                                                                                                                          |
| Ackley6d     | $f(\mathbf{x}) = -20 \exp \left( -0.2 \cdot \sqrt{\frac{1}{6} \sum_{i=1}^6 x_i^2} \right) - \exp \left( \frac{1}{6} \sum_{i=1}^6 \cos(2\pi x_i) \right) + 20 + \exp(1)$ |
| Rosenbrock6d | $f(\mathbf{x}) = \sum_{i=1}^6 (0.4 - x_i)^2 + 20 \sum_{i=1}^5 (x_i - x_{i+1})^2$                                                                                        |

**Table 2**Specifications of the original GE-E<sup>3</sup> LPT blade and the ultra-highly loaded blade.

|                                     | GE-E <sup>3</sup> LPT blade | Ultra high-loaded blade |
|-------------------------------------|-----------------------------|-------------------------|
| Inlet flow angle / °                | 49.3                        | 49.3                    |
| Inlet total pressure / Pa           | 319339.2                    | 319339.2                |
| Inlet total temperature / K         | 1161                        | 1161                    |
| Outlet static pressure / Pa         | 245618                      | 245618                  |
| Axial chord length / mm             | 26.6                        | 26.6                    |
| Pitch / mm                          | 25.006                      | 30.0072                 |
| Outlet flow angle / °               | -64.18                      | -60.86                  |
| Total pressure loss coefficient / % | 4.22                        | 12.99                   |

## 5. Engineering optimization

To show the effectiveness of proposed KT-ASO, an aerodynamic optimization of GE-E<sup>3</sup> turbine blade is carried out in this section.

### 5.1. Problem description

To meet the requirement of higher thrust-weight ratio and low fuel consumption of modern aeroengines, people resort to reducing the number of blades in a cascade of low-pressure turbine (LPT) [14]. Nonetheless, the aerodynamic loss in highly loaded LPT blades can be increased significantly [44]. Then, the design of highly loaded turbine blade with low aerodynamic loss has drawn wide attentions.

As a preliminary study, the mid-span section profile of the rotor blade of the second stage of GE-E<sup>3</sup> LPT [45] (labeled as rb23) is taken as the reference design. The aerodynamic characteristics of highly loaded blade are investigated by simply reducing the number of referenced blades from 120 to 100, while the inlet and outlet boundary conditions of the rotor blade remained. Table 2 shows the geometric specifications and working conditions of the original GE-E<sup>3</sup> LPT blade and the highly loaded blade, and Fig. 10 shows the distribution of isentropic Mach number at blade surface and the streamlines near the trailing edge of the blades. While increasing the loading of the blade by reducing the number of blades from 120 to 100, an obvious separation bubble is observed at the trailing edge of blade, and the total loss coefficient  $C_{pt}$  increases from 4.22% to 12.99%, as shown in Table 2.

In view of the above, the target optimization problem is set as minimizing the total loss coefficient  $C_{pt}$ , with constraint on the mass flow rate of the highly loaded blade. In the meantime, considering the mechanical strength of the blade, the minimum radius of the trailing edge (denoted by  $r_{min}$ ) of the optimized blade is confined to be not thinner than the referenced design. The related mathematical optimization expression for the target optimization problem is shown as follows:

$$\begin{aligned} \min f(\mathbf{z}) &= \min C_{pt}(\mathbf{z}), \\ \text{s.t. } 0.98 \cdot m_{ref}^{ultra} &\leq m(\mathbf{z}) \leq 1.02 \cdot m_{ref}^{ultra}, \\ r_{min}(\mathbf{z}) &\geq r_{min}^{ref} \end{aligned} \quad (13)$$

where,  $m_{ref}^{ultra}$  denotes the mass flow rate of the ultra-highly loaded blade shown in Table 2.

To mimic the situation of transfer optimization, the optimization of GE-E<sup>3</sup> LPT blade at its original blade loading is assumed to have been

finished. The optimization model of the completed task is shown as follows:

$$\begin{aligned} \min f(\mathbf{z}) &= \min C_{pt}(\mathbf{z}), \\ \text{s.t. } 0.98 \cdot m_{ref} &\leq m(\mathbf{z}) \leq 1.02 \cdot m_{ref}, \\ r_{min}(\mathbf{z}) &\geq r_{min}^{ref} \end{aligned} \quad (14)$$

where,  $m_{ref}$  denotes the mass flow rate of the GE-E<sup>3</sup> LPT blade shown in Table 2.

Similar to the numerical experiments in Section 4.1, Bayesian optimizations are carried out in advance for Eq. (14) for simulating transfer source samples. Specifically, a total of 1400 intermediate samples from 10 individual runs are collected, and the top 400 results with lower  $C_{pt}$  are taken out as candidates. For each run of KT-ASO for Eq. (13), 40 samples are randomly drawn from the candidates, and selected as the transfer source samples  $\{\mathbf{x}_S, y_S\}_m$ . The effect of  $\{\mathbf{x}_S, y_S\}_m$  will be discussed in Section 5.3.

### 5.2. Neuron network architecture and training loss of VAE for blade parameterization

Before detailed discussion of the effect of including  $\{\mathbf{x}_S, y_S\}_m$  as part of the training samples for VAE training, the neuron network architecture and the training loss of VAE for blade parameterization is first illustrated, as shown in Fig. 11, and the detailed settings of each layer are listed in Table 3.

As shown in Fig. 11, the input and output of VAE are the blade profiles that are depicted by a vector  $\mathbf{x}$  consisted of 202 two-dimensional coordinate scatter points. Meanwhile, the dimension of the VAE latent space is set to be 10. Considering the smoothness of blade profile, a FFD layer is embedded at the end of VAE decoder. Wherein, the control point of FFD is a  $6 \times 5$  grid. Each control point has 2 degrees of freedom, except those on the leading and trailing edges. The total number of variables of FFD active control points is 50.

To avoid generating abnormal blade profiles, a physical constraint is imposed to the blade trailing edge, with the following equation:

$$L_p = E \left( \left( \max \left\{ 0, \frac{r_{min}^{ref} - r_{min}}{r_{min}^{ref}} \right\} \right)^2 \right) \quad (15)$$

In other words, in order to avoid the distortion of blade trailing edge, the blade profiles generated by VAE are confined to be not thinner than the referenced GE-E<sup>3</sup> LPT blade. In the meantime, we set  $\lambda_1 = 1, \lambda_2 = 500000$  in Eq. (8) when training the VAE model for turbine blade parameterization.

### 5.3. Data preparation and analysis of the effect of source samples for the VAE training

While the generative model based parameterization method has been shown to be more efficient than the conventional parameterization methods in the previous studies, the selection of training dataset for generative model can greatly influence its effectiveness. The geometric parameterization method shown in Fig. 2 is used to generate the training dataset for the VAE training.



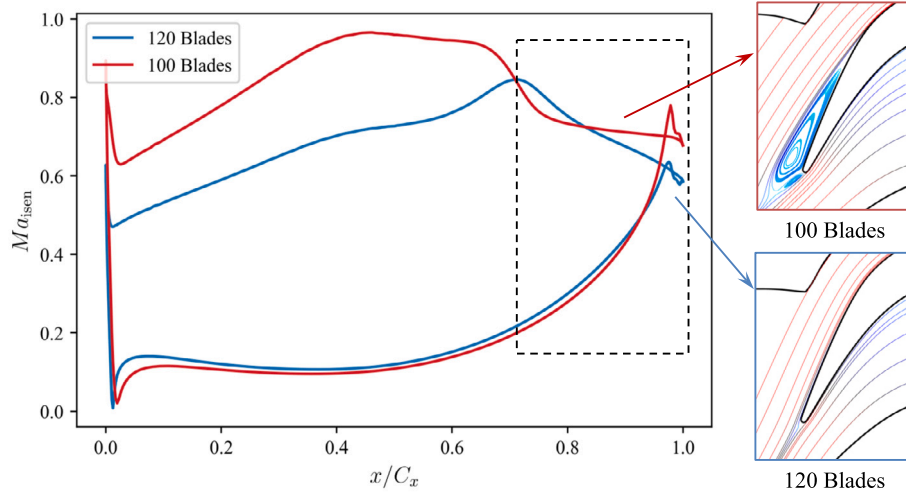


Fig. 10. The isentropic Mach number distribution at blade surface and streamlines near the blade trailing edge.

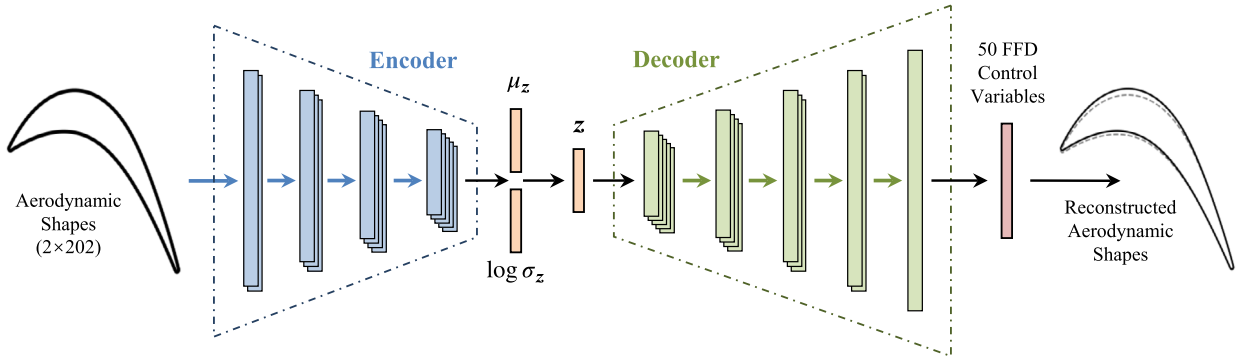


Fig. 11. Architecture of VAE for turbine blade parameterization.

Table 3

Neural network architecture of VAE model for the turbine blade parameterization.

| Encoder                                                   | Decoder                                          |
|-----------------------------------------------------------|--------------------------------------------------|
| Input $2 \times 202$ aerodynamic shape                    | Input $\in \mathbb{R}^{10}$                      |
| Downsample $8 \times 101$                                 | FC $512 \times 12$ , LeakyReLU 0.2               |
| Downsample $32 \times 51$                                 | Upsample $128 \times 24$                         |
| Downsample $128 \times 26$                                | Upsample $32 \times 48$                          |
| Downsample $512 \times 13$                                | Upsample $8 \times 96$                           |
| FC $10 \times 2$ , output for $\mu_z$ and $\log \sigma_z$ | Upsample $2 \times 192$                          |
|                                                           | FC 50, sigmoid, output for FFD control variables |

Specifically, the section profiles of GE-E3 LPT blades are taken as the reference designs, which are first fitted by the geometric parameterization method. Then, by varying the values of inlet geometric angle  $\beta_1$ , upper wedge angle  $\epsilon_{up}$ , outlet airflow angle  $\beta_2$ , outlet deflection angle  $\delta_{out}$ , angle of circle center line  $\gamma$  and axial chord length  $C_{ax}$  by  $\pm 10\%$  with respect to the datum value of the related reference profile, 1010 blade profiles without CFD simulations are generated, denoted by  $X_U$ , which are scaled to the uniform size and visualized by t-SNE plot in Fig. 12. To be noted, t-SNE is short for t-distributed stochastic neighbor embedding, which is a clustering technique that is used to visualize the high-dimensional dataset in two- or three-dimensional space. Each scatter point represents a sample profile, and the related profiles of the adjacent scatters are in similar shape, while the blade profiles of the scatters that are far from each other are quite different. As shown in Fig. 12, the six profiles of the scatters that are distributed in different regions look different, indicating that the prepared dataset has sufficient shape variability.

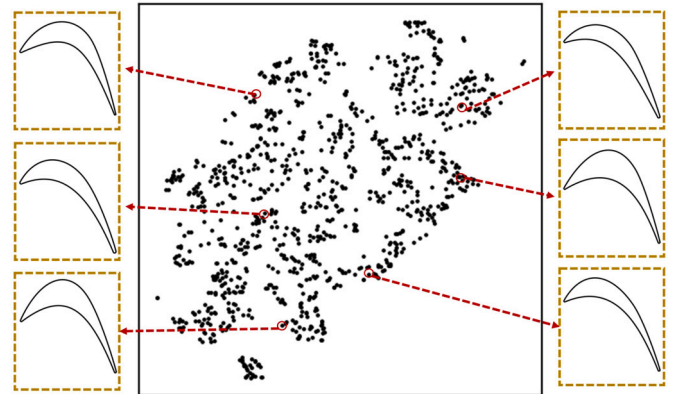


Fig. 12. Visualization of the unevaluated blade profiles using t-SNE.

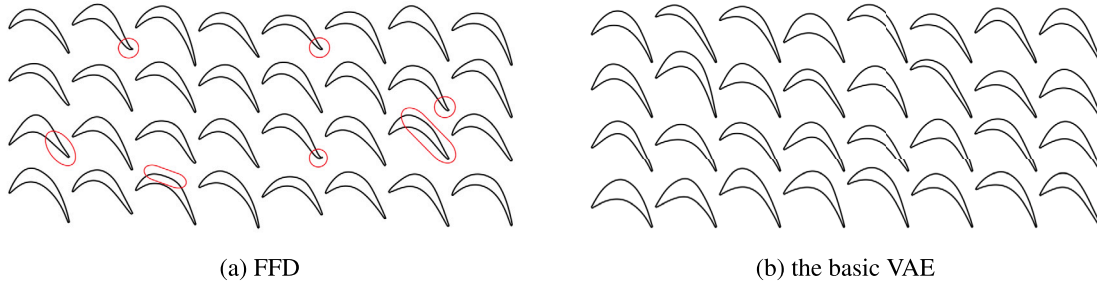


Fig. 13. Random turbine blades generated by different parameterizations, significant distortions on generated blades are marked by circles.

Table 4

Accuracy comparison of the basic and sample-weighted VAE for blade reparameterization.

|                                                                    | Reconstruction error (normalized MSE) |
|--------------------------------------------------------------------|---------------------------------------|
| Reparameterization and reconstruction with the basic VAE           | $(8.610 \pm 3.168) \times 10^{-3}$    |
| Reparameterization and reconstruction with the sample-weighted VAE | $(1.986 \pm 1.572) \times 10^{-4}$    |

For comparison purpose, a VAE model is first trained with the 1010 unevaluated blade profiles (i.e.,  $X_U$ ), and an sample-weighted VAE is trained by using the training scheme in Section 3.2 with both  $X_U$  and  $\{x_S\}_m$ , where the ratio as  $n^{(S)}/n^{(T)}$  in Eq. (9) is set to be 1. In other words, the same number of samples are randomly drawn from  $X_U$  and  $\{x_S\}_m$ , respectively, in each batch for the training of sample-weighted VAE. To show the sample variability of the basic VAE, it is first compared to the conventional blade parameterization method FFD (short for free-form deformation), as shown in Fig. 13. It can be observed that the sample profiles generated by FFD look more similar to each other, and a portion of sample profiles of FFD is distorted near the trailing edge, which are marked by red circles. In contrast, the profiles generated by the standard VAE model look more diverse, and almost no distorted samples are observed. These results are consistent with the statements of Chen et al. [7,46] and Li et al. [8].

To analyze the effect of transfer source samples on the VAE training, Fig. 14 presents the sample distributions of the learnt latent design space, and Fig. 15 compares the reparameterization accuracy of the basic and sample-weighted VAE. As shown in Fig. 14(a), the 40 optimized solutions of the related completed tasks (i.e.,  $\{x_S\}_m$ , marked by the dashed circle) lie in the marginal area of the learnt design space of the basic VAE. In other words, the samples in  $\{x_S\}_m$  are rarely observed in the design space of the basic VAE. As a consequence, when using the basic VAE to reparameterize  $\{x_S\}_m$  into the VAE design space, obvious gaps between the original and reparameterized blade profile are observed, as shown in Fig. 15(a).

In contrast, by including  $\{x_S\}_m$  as part of the dataset and training the VAE model with the strategy illustrated in Section 3.2,  $\{x_S\}_m$  are shown to be well covered by the learnt design space of sample-weighted VAE. Then, when reparameterizing  $\{x_S\}_m$  to the VAE design space, the reparameterized blade profiles and the original one are almost exactly the same, as shown in Fig. 15(b), and the reconstruction error of sample-weighted VAE is shown to be reduced by more than one order of magnitude, as shown in Table 4. In other words, though  $\{x_S\}_m$  only accounts for a very small portion of the full training dataset, their features have been properly learnt by using our proposed sample-weighted training strategy. Moreover, as the features of the optimal solutions of the related completed task and target problem are similar, the well covering of  $\{x_S\}_m$  also means that the proportion of high-scoring solutions, which can substantially improve the design performance of target problem, has been increased significantly. Then, as will be seen in Section 5.5, such improvement facilitates the optimization search of target problem.

#### 5.4. Settings of the compared ASO procedures

Before detailed discussion on the optimization results of turbine blades, the settings of VAE-based optimization process and the compared ASO procedures are first illustrated. Specifically, in addition to carrying out optimization with our proposed KT-ASO as shown in Fig. 9, the following ASO procedures are used as comparing baselines.

To show the benefit of sample-weighted VAE which includes the source samples as part of the training set, the optimization of turbine blades is carried out over the latent space of both basic VAE and sample-weighted VAE. The standard BO algorithm shown in Section 2.4 is used to guide the optimization search, and the related optimization results are labeled as VAE-BO and VAE-W-BO, respectively. Then, the benefit of using multi-fidelity surrogate to gain knowledge from the source samples can be seen by comparing the optimization results of VAE-W-BO and KT-ASO.

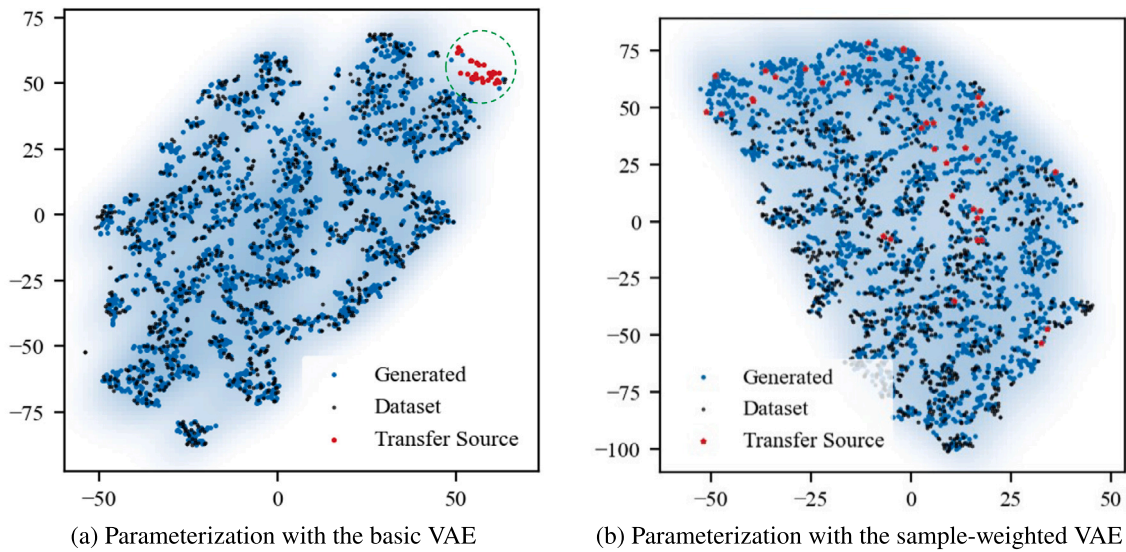
To verify the advantage of VAE-based parameterization method over the conventional parameterization method, a standard ASO is also conducted with FFD, where the mid-span section profile of the GE-E<sup>3</sup> rotor blade (i.e., rb23) is selected as the reference design for the FFD-based parameterization. The standard BO algorithm shown in Section 2.4 is used to guide the optimization search, the related optimization results are labeled as FFD-BO.

For comparison with recent similar studies on generative model-based parameterization and optimization, we established a generative adversarial net (GAN) model for blade parameterization by reusing the network structure of existing encoder and decoder, and training the model with modified scheme [9]. Theoretically, the generation capability of GAN is stronger than VAE, however its training is more difficult. Also, the availability of sample reparameterization and sample-weighted model training on GAN require further study. In this work, a standard Bayesian optimization similar to FFD-BO is carried out using GAN proposed above for parameterization, labeled as GAN-BO.

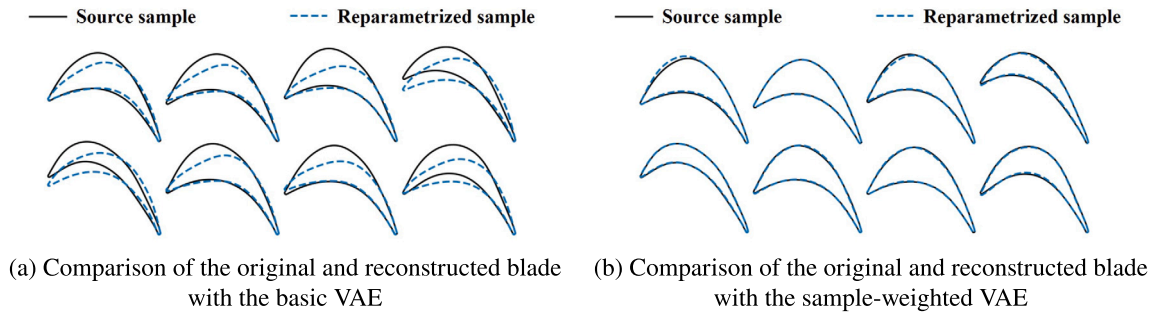
Additionally, for fair comparison, the sample budget of all ASO procedures is set to be 140, i.e., the optimization by using FFD-BO, GAN-BO, VAE-BO, VAE-W-BO and KT-ASO is terminated when 140 CFD evaluations have been conducted for the target optimization problem.

#### 5.5. Results discussion

Fig. 16 shows the optimization results of the compared ASO procedures over 10 runs. Our proposed KT-ASO is shown to achieve the best solutions with fastest convergence rate. Specifically, when comparing to FFD-BO, the three VAE-based ASO procedures attain significantly better solutions. Such phenomenon is consistent with the previous studies [7,8], and the reason behind can be explained as follows. While the



**Fig. 14.** Visualization of the shape variability of blade parameterization design space built by the basic VAE and sample-weighted VAE, where the shaded area denotes the sample distributions of the VAE learnt design space.



**Fig. 15.** Comparison of the reparameterization accuracy of basic VAE and sample-weighted VAE for turbine blade design.

design space of FFD-BO is built by adjusting profiles around a single referenced profile, the design space of VAE-BO, VAE-W-BO and KT-ASO is learnt with at least 1010 different blade profiles shown in Fig. 12. In other words, the sample variability in VAE-based design space can be much better than that of FFD-based design space (as illustrated in Fig. 13), which means the proportion of high-scoring solution candidate of VAE-based design spaces is much higher than that of FFD-based one. Therefore, both the starting points and final solutions of VAE-based ASO procedures are shown significantly better than that of FFD-BO. Furthermore, the average convergence speed of VAE-BO slightly outperforms the speed of GAN-BO, and VAE-BO has a smaller variance in result's performance. The result could be attributed to the difficulty in training the GAN model. The observation on convergence history proves that the generation quality of VAE is roughly at the same level of GAN or similar deep generative models.

When comparing VAE-BO and VAE-W-BO, VAE-W-BO achieves much better final solutions with smaller result variance. By referring to Fig. 14, the reason behind can be explained as follows. That is, compared to the basic VAE, the source samples of the completed tasks are much better covered by the learnt design space of sample-weighted VAE. In the meantime, as seen in Fig. 17, the features of the optimal solutions of the related completed task and the target problem are similar, which are located in similar regions of t-SNE plot. Hence, the better coverage of source samples also means the proportion of high-scoring solutions in the design space of sample-weighted VAE is much higher than that of basic VAE, which can facilitate the optimization search, and thus making both the starting point and the final solutions of VAE-W-BO are much better than those of VAE-BO.

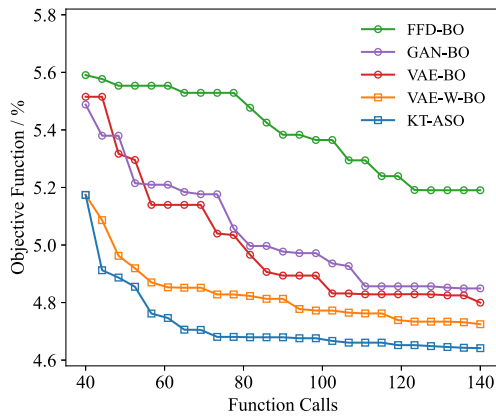
**Table 5**

The overall performance comparison between the benchmark and the median of the optimal blade designs.

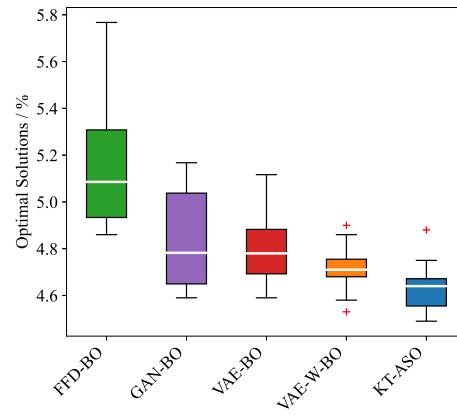
| Blade           | $C_{pt} / \%$ | $\dot{m} / \text{kg} \cdot \text{s}^{-1} \cdot \text{m}^{-1}$ | Outlet flow angle / $^\circ$ |
|-----------------|---------------|---------------------------------------------------------------|------------------------------|
| Benchmark(rb23) | 12.99         | 475.02                                                        | -60.86                       |
| FFD-BO          | 4.96          | 472.45                                                        | -61.04                       |
| GAN-BO          | 4.86          | 474.47                                                        | -60.95                       |
| VAE-BO          | 4.85          | 470.18                                                        | -61.25                       |
| VAE-W-BO        | 4.75          | 468.78                                                        | -61.35                       |
| KT-ASO          | 4.52          | 473.32                                                        | -61.06                       |

When comparing VAE-W-BO and KT-ASO, KT-ASO achieves even better final solutions with much faster convergence rate. Different from VAE-W-BO which uses target samples alone in the optimization search, KT-ASO uses multi-fidelity surrogate to instantly gain knowledge from the source samples to accelerate the optimization search. In other words, the involvement of source samples in KT-ASO can help the algorithm to arrive at the small neighborhood of the target optimal solution much more quickly. Hence, both the convergence rate and final solutions of KT-ASO are better than those of VAE-W-BO.

With the above, the better performance of KT-ASO can be summarized as follows. First, the including of source samples in the training set of VAE helps to increase the proportion of high-scoring solution candidates in the searching space. Second, by directly and constantly leveraging knowledge from the source samples to guide the optimization search, the use of multi-fidelity surrogate in KT-ASO can help the algorithm to arrive at the vicinity of the target optimal solution much more quickly.

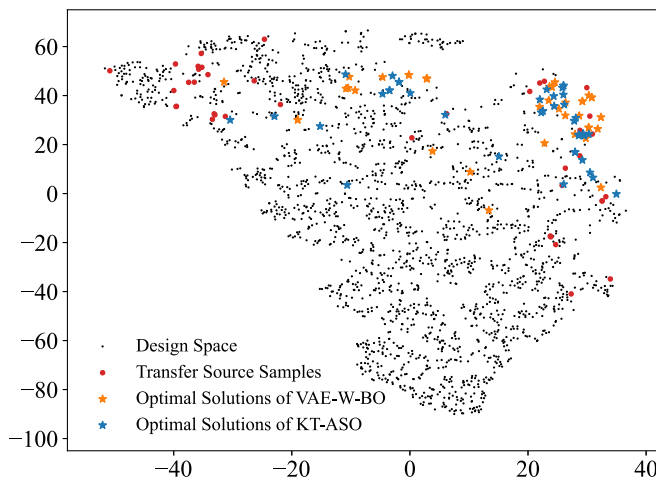


(a) Mean convergence curve over 10 runs



(b) Distributions of the final optimal solutions

**Fig. 16.** The mean convergence curve and the distributions of the final optimal solutions over 10 runs of the turbine blade optimization.



**Fig. 17.** t-SNE plot of the blade transfer source samples and the optimal solutions of VAE-W-BO and KT-ASO over 10 runs.

Furthermore, the median of the optimal solutions of the compared ASO procedures is picked out for aerodynamic analysis. Table 5 lists the overall performance of the median optimal solutions, and Fig. 18 shows the static entropy contours of the median optimal solutions. While meeting the constraints on mass flow rate and the outlet flow angle as shown in Eq. (13), all optimized solutions achieve better loss coefficient. As shown in Fig. 18, the entropy values at the blade trailing edge of optimized solutions of FFD-BO, GAN-BO, VAE-BO, VAE-W-BO and KT-ASO are significantly reduced, indicating that the separation bubble of the optimized solutions is greatly reduced. Moreover, area of high-entropy loss region of KT-ASO is the smallest. With the above, the effectiveness of the proposed knowledge transfer accelerated ASO process, i.e., KT-ASO, is well demonstrated.

## 6. Conclusion

Inspired by the concept of transfer learning, a knowledge transfer accelerated aerodynamic shape optimization (ASO) procedure is proposed, namely KT-ASO, which makes use of the samples from related completed tasks (also known as source samples) to accelerate the optimization progress of the target turbine design problem. Through the test on aerodynamic shape optimization of a highly loaded turbine blade, some conclusions are drawn as follows:

(1) The proposed sample-weighted training strategy can effectively learn the distinct features of the samples that accounts for a very small proportion of the whole training dataset.

(2) Since the features of the optimal solutions of the completed task and target problem are very similar, the including of the source samples with our proposed sample-weighted training strategy can help to increase the proportion of high-scoring solution candidates in the learnt target design space, and thus increasing the probability to attain substantially better solutions.

(3) The including of source samples can also help to increase the accuracy of reparameterizing the source samples in the target design space. Upon that, the useful information of optimized solutions of the completed tasks can be accurately extracted by our proposed multi-fidelity surrogate, and thereby helping the algorithm to more quickly arrive at the small neighborhood of the true optimal of the target problem. As a consequence, better solutions with faster convergence rate are observed when using KT-ASO to carry out the turbine blade design.

Also note that the framework of our proposed KT-ASO process is quite general, which can be further extended for the 3D multidisciplinary design optimization of turbine blades and other turbomachinery applications.

## CRediT authorship contribution statement

**Zhendong Guo:** Writing – original draft. **Cunxi Li:** Investigation. **Yun Chen:** Writing – review & editing. **Liming Song:** Writing – review & editing. **Jun Li:** Writing – review & editing, Conceptualization. **Zhenping Feng:** Writing – review & editing, Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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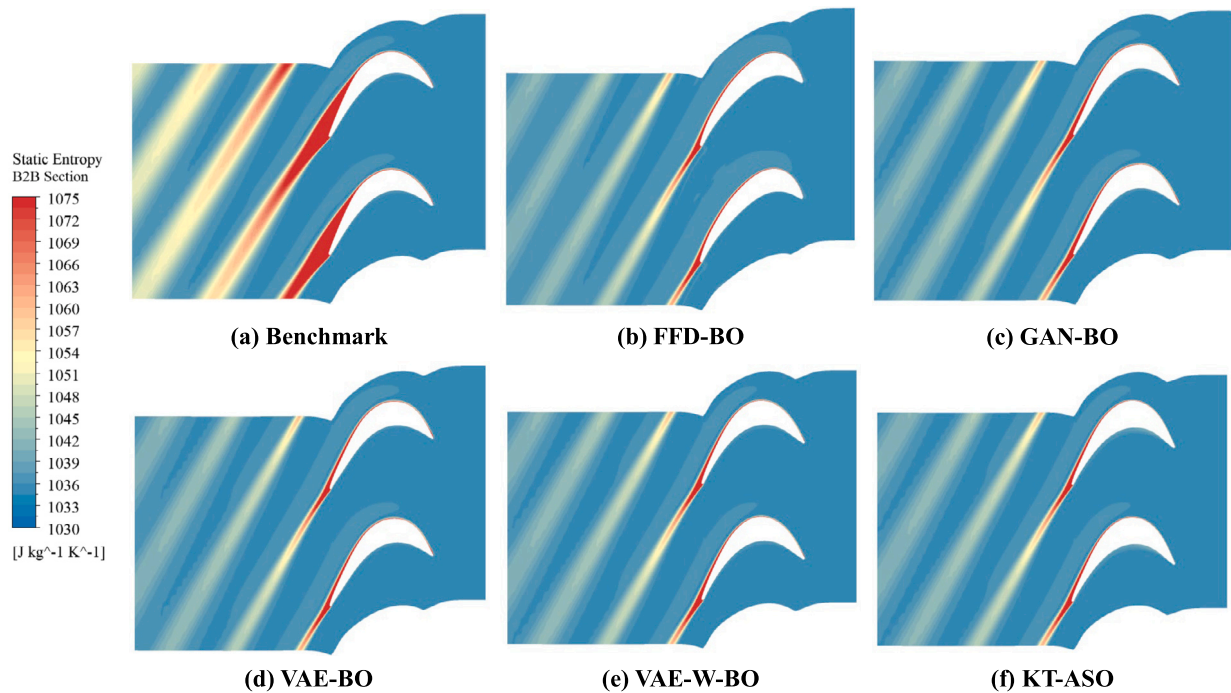


Fig. 18. Comparison of the static entropy distributions of the benchmark and the median of optimal design.

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